

Operations

Week 1 – Live Class

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About me 1/2

Gerardo Berbeglia – Associate Professor of Operations Management

Education

- Ph.D. (Operations Research), Université de Montréal, Canada
- B.Sc and M.Sc in Computer Science, Universidad de Buenos Aires, Argentina

About me 2/2

Professional Experience

- 2009-2011 Consultant – Senior Scientist, ExPretio Technologies Inc
 - Research: pricing optimization, customer behaviour modelling.
 - Consulting: Air Canada (canada), Amtrak (US), SNCF (France), Thalys (Belgium).
- 2011 Natural Sciences and Engineering Research Council of Canada (NSERC) postdoctoral fellow, McGill University
 - Algorithmic game theory
 - Pricing optimisation problems in durable goods monopolies
- 2012 – today: MBS
 - MBA (Operations)
 - MBusA (Optimisation and Supply Chain Analytics)
 - Research: Transport Optimisation, Revenue Management, Pricing Optimisation, Effects of Social Influence
 - Consulting: Airlines

OBJECTIVES

- To understand business processes and challenges;
- Learn tools to perform a structured analysis of existing business processes
- Prepare managers to use the results to continuously improve the firm's operational performance.
- Effective operations management is about effectively matching supply with demand.

The Economic Consequences of Supply-Demand Mismatch

	Air travel	Emergency room	Retailing	Iron ore plant	Pacemakers
Supply	Seats on flights	Medical service	Consumer electronics	Iron ore	Medical equipment
Demand	Travel preferences of the population	Urgent need for medical service	Consumers buying a new video system	Steel mills	Heart surgeon requires pacemaker at exact time and location
Supply exceeds demand	Empty seats	Doctors, nurses, and infrastructure are under-utilized	High inventory costs; few inventory turns	Prices fall	Pacemaker sits in inventory
Demand exceeds supply	Premature full booking	Crowding and delays in the ER, potential diversion of ambulances	Foregone profit opportunity; consumer dissatisfaction	Prices rise	Foregone profit (typically not associated with medical risk)
Actions to match supply and demand	Optimal number of flights - Dynamic pricing optimization	Right staffing for the predicted demand; priorities	Forecasting; quick response	If prices fall too low, production facility is shut down	Distribution system holding pacemakers at various locations
Managerial importance	Small variations in the load factor produce a strong impact on profits	Delays in treatment or transfer have been linked to death;	Per unit inventory costs for consumer electronics retailing commonly exceed net profits.	Prices are so competitive that the primary emphasis is on reducing the cost of supply	Most products (valued \$20k) spend 4-5 months waiting in a trunk of a sales person before being used

Practice questions

Question 1

0 pts

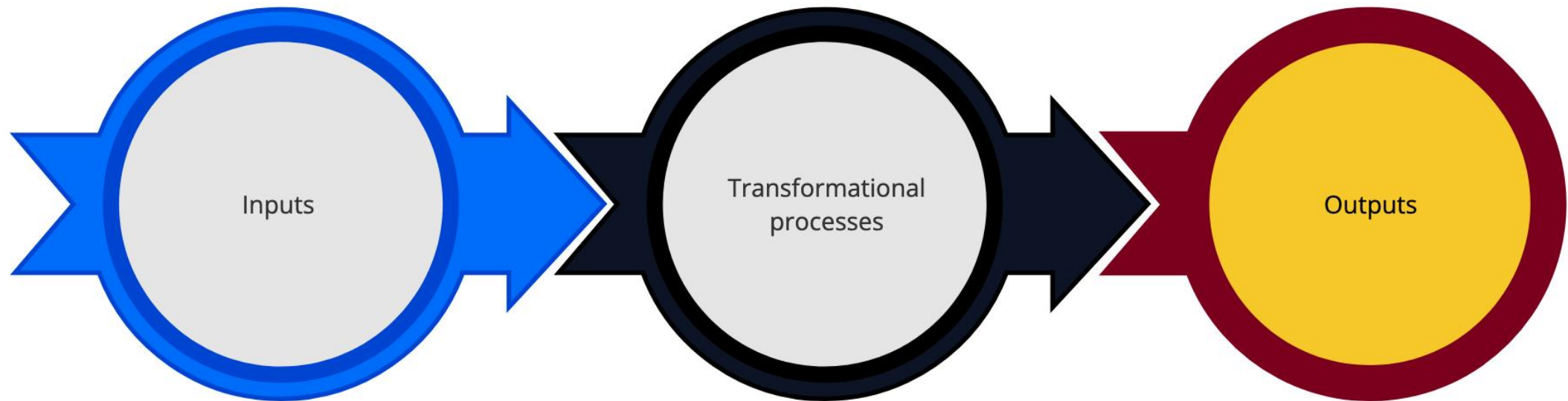
Determine whether the statement is true or false.

A company that finds it difficult to match demand with supply can always solve the problem by adjusting its prices.

☐ True

☐ False

Input-transformation-output model



A BAKERY

Inputs

- Buildings, equipment, food ingredients, packaging materials, energy, and people.

Transforming processes

- Mixing, proofing, baking, packaging, maintenance, distribution.

Outputs

- Range of packed bakery products delivered to warehouses and retail outlets that are fresh and in line with the needs of the customers.



AIR PASSENGER TRANSPORT

Inputs

- Airports, Booking systems, aircrafts, aircrews, ground staff, fuel, food.

Transformational processes

- Passenger reservations, flight schedules, aircraft maintenance, aircraft cleaning and refueling, meals and crew scheduling, boarding procedures, in-flight procedures, baggage claim

Outputs

- Customers booked on appropriate flights in terms of timing and convenience
- Customers progressed through pre-boarding phase of the service delivery system
- Customers transported safely and on time to chosen destinations.



Operations Strategy

Broad classification based on: Cost, speed, flexibility, quality

Cost: Strives for being the lowest-cost producers in their industry. Example: Walmart

Speed or responsiveness: Priority is given to reduce time to develop, produce and/or delivery. Example: Amazon or Zara.

Flexibility: Firm with high flexibility can quickly adapt to changes in customer demand or market conditions. Example: Toyota

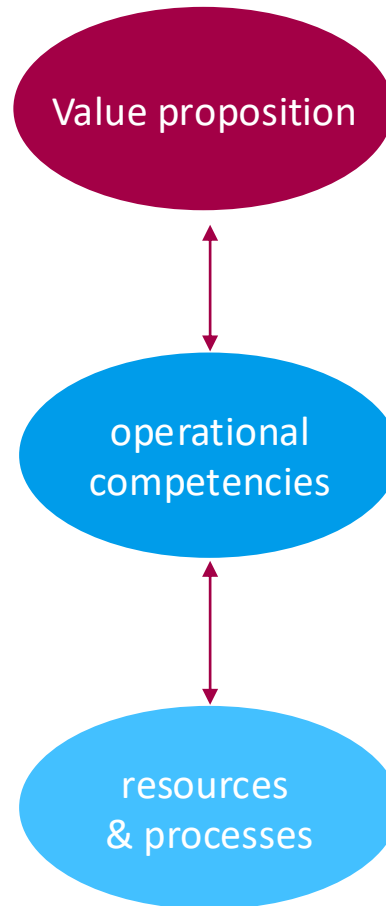
Quality: Aims at delivering the highest-value product/service within industry. Example: Rolex

Value Proposition

Value proposition: Firm's promise to deliver specific value to its customers, distinguishing itself from competitors.

- What consumer problems are we trying to address? What consumer desires are we trying to satisfy?
- How would our offerings solve these problems or needs?
- What benefits do customers get from buying our products or services?
- Why would they choose us and not the competitors?

Designing an Operations Strategy



What is our strategic position: how do we compete & provide value in the market?

- What is the *value* proposition to our customers?

Given our strategic position, what must operations do particularly well?

- Which *competencies* must ops develop?

Given needed competencies, how should operations processes be structured to develop competencies that support strategy?

- Process choice (*structure*) and management

Operations and Competitive Advantage - Differentiation

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Online



Zara: Spanish clothing and accessories retailer (€18 billion sales in 2018)

- Competitors (Gap, H&M) outsource production to countries around the world, many of them in low cost **Asia**
- 80% of Zara's production in **Europe** (within a small radius of its headquarters in Spain)
- Rather than concentrating on forecasting accurately, developed its business to **react swiftly** to emerging trends
- Needs just **two weeks** to develop a new product and get it to stores, compared to the six-month industry average,
- Launches around 10,000 new designs each year.

Operations and Competitive Advantage - Differentiation



Southwest Airlines: low-cost carrier founded in 1969.

- Business model: focus on short-haul point-to-point flights
- Market: domestic U.S.
- Standardised fleet: Boeing 737
- Most competitors: Hub-and-Spoke

Discussion: What are the benefits of a standardised fleet? What are the benefits of Point-to-Point flights compared with Hub-and-Spoke?

And the Cons?

The Carbon Footprint of Firms



Scope 1: Direct emissions from sources owned or controlled by the firm.

Scope 2: Indirect emissions generated from purchased or acquired energy (electricity, heating, cooling, etc).

Scope 3: All other indirect emissions occurring in a company's value chain.

Sustainability in Operations: Why?



- Risk management
- Cost savings
- Brand building
- Access to capital
- Employee engagement

Sustainability Strategies in Operations:



How?

- Energy efficiency and renewable energy
- Sustainable supply chain management
- Sustainable product design
- Waste reduction and resource efficiency

Ikea's sustainability strategies



- Circular products: Design for adaptability
- Forest ownership
- Buy-back & resell
- Furniture rental model
- Product servicing efforts

Ikea's sustainability strategies



- Which of IKEA's sustainability strategies do you believe is the most difficult to implement, and why?
- Which of these strategies do you believe has the most positive impact on IKEA's customers?
- Can you think of other organisations that could apply similar sustainability strategies?

PROCESS FLOW DIAGRAM:

Components



Storage areas or queues



Tasks or Operations



Flow of material or customers

THREE MEASURES OF PROCESS PERFORMANCE

Cycle Time (time/unit)

- Average time between the output of successive units (e.g. 5 min/unit)
- **Flow rate or throughput rate** is the inverse of the cycle time (e.g. $1/5$ units/min = 0.2 units/min)

Flow time or manufacturing lead time or service lead time

- The average amount of time it takes a unit to get through the process (includes waiting time).

Inventory or Work in process (WIP)

- Average number of units in the process at any point in time (includes the units waiting to be processed).

Examples of flow time, flow rate and inventory

	U.S. immigration department	MBA program	PC manufact- urer	Champagne industry
Flow unit	Application for immigration benefit	MBA student	Computers	Bottles of champagne
Flow rate (R)	Approved or rejected cases: 6.3 million per year	600 students per year	5,000 units/day	260 million bottles per year
Flow time (T)	Average processing time: 7.6 months	2 years	10 days	Average time in cellar: 3.46 years
Inventory (I)	Pending cases: 4 million cases	1,200 students	50,000 computers	900 million bottles

Practice questions

Question 2

0 pts

Choose the correct answer from the options provided.

A consulting firm is hired to study the security process required for each passenger at an airport. The flow rate in such a process is most likely to be:

- ☐ Number of passengers.
- ☐ Average duration of flights.
- ☐ Number of passengers per flight.
- ☐ Number of flights.
- ☐ Number of passengers per hour.

HOW ARE THESE MEASURES LINKED TO COST AND PROFIT?



These measures have a direct impact on costs and on profits.

- Shorter flow times:
 - higher responsiveness
 - additional unit sales and/or higher prices
- Lower inventory:
 - lower working capital requirements
- Higher flow rate:
 - Higher revenues

Capacity vs Flow rate

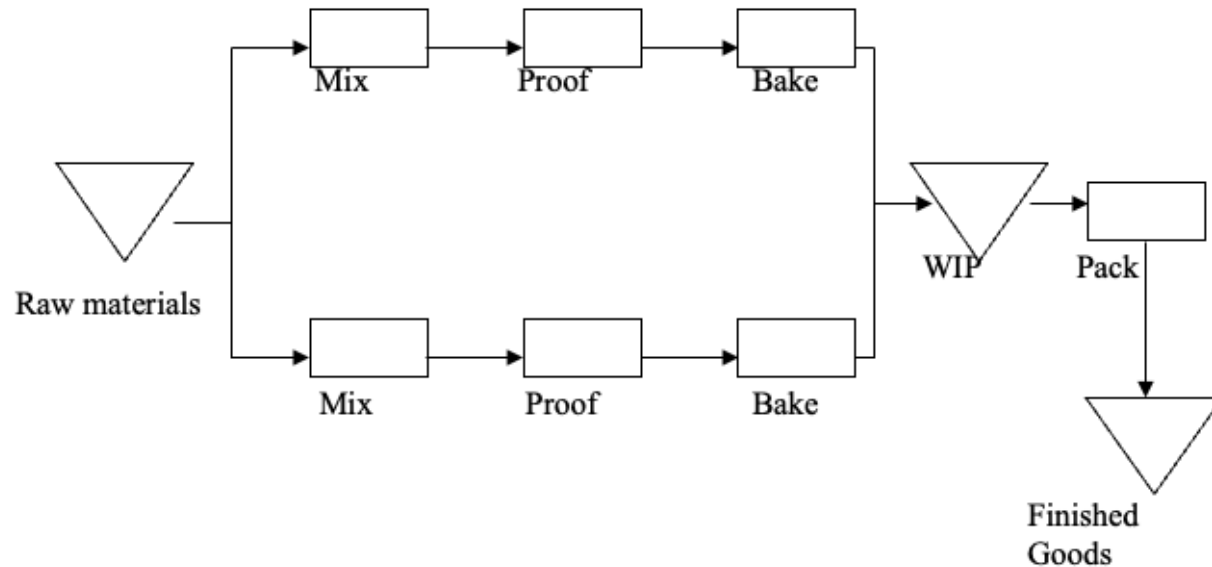
Process Capacity: maximum rate at which the process can run.

Flow rate: rate at which the process is running.

How do you identify a bottleneck?

Have you ever faced a long delay due to an unexpected bottleneck?

Rise 'n' Shine



1 batch = 100 loaves

Processing times:

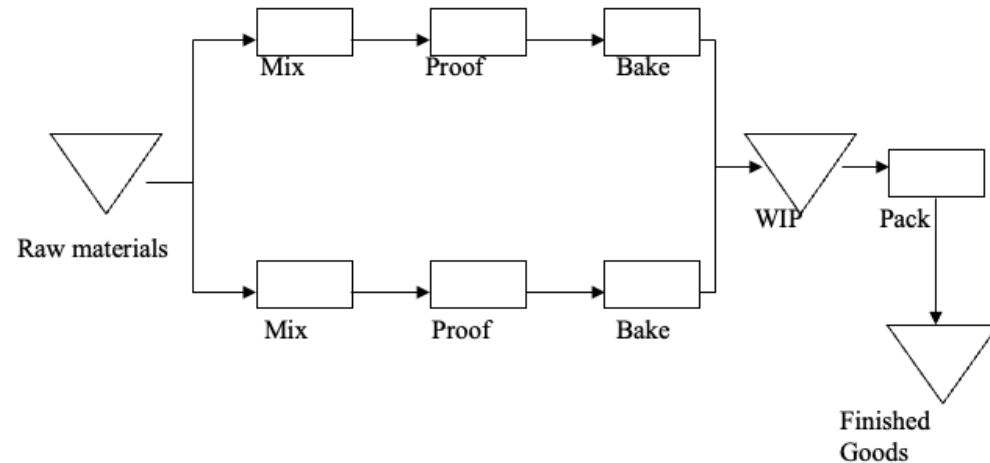
For mix operation --- $\frac{3}{4}$ hour per batch for each line

For proofing --- $\frac{3}{4}$ hour per batch for each line

For baking --- 1 hour per batch for each line

For packing --- $\frac{3}{4}$ hour per batch

Rise 'n' Shine

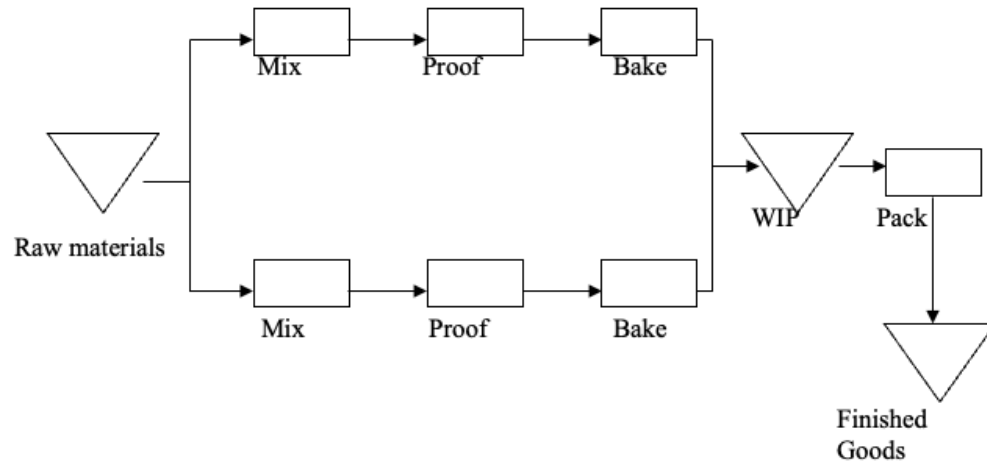


Step	Processing time per batch (each line)	Capacity per line (batches per hour)	Total capacity
Mixing	90 min per batch	1 batch/1.5 hr = $\frac{2}{3}$ batch per hour	$2 * \frac{2}{3} = \frac{4}{3}$ batch per hour
Proofing	90 min per batch	1 batch/1.5 hr = $\frac{2}{3}$ batch per hour	$2 * \frac{2}{3} = \frac{4}{3}$ batch per hour
Baking	120 min per batch	1 batch/2 hr = $\frac{1}{2}$ batch per hour	$2 * \frac{1}{2} = 1$ batch per hour
Packing	90 min per batch	1 batch/ 1.5 hr = $\frac{2}{3}$ batch per hour	$\frac{2}{3}$ batch per hour

Capacity: $\frac{2}{3}$ batch per hour
Bottleneck: Packing step

Rise 'n' Shine

Improved packaging technology



Step	Processing time per batch (each line)	Capacity per line (batches per hour)	Total capacity
Mixing	90 min per batch	1 batch/1.5 hr = 2/3 batch per hour	2 * 2/3 = 4/3 batch per hour
Proofing	90 min per batch	1 batch/1.5 hr = 2/3 batch per hour	2 * 2/3 = 4/3 batch per hour
Baking	120 min per batch	1 batch/2 hr = 1/2 batch per hour	2 * 1/2 = 1 batch / hr
New Packing step	60 min per batch	1 batch/ hr	1 batch / hr

New capacity: 1 batch per hour!
Bottlenecks: Packing step and Baking step

Bottleneck in a Car Wash

You manage an automated car wash with the following sequence of steps that every car must go through in the same order:

1. Pre-Wash Spray (2 minutes per car)
2. Soap Application (3 minutes per car)
3. Rinse (2 minutes per car)
4. Drying (5 minutes per car)

On a busy weekend afternoon, the car wash is seeing long queues, and customers are complaining about wait times. You suspect a bottleneck is causing the delays.

Bottleneck in a Car Wash

1. Pre-Wash Spray (2 minutes per car)
2. Soap Application (3 minutes per car)
3. Rinse (2 minutes per car)
4. Drying (5 minutes per car)

Discussion questions:

- 1) Where is the bottleneck?
- 2) How does this bottleneck impact the overall flow of cars?
- 3) What solutions can be implemented to reduce delays?

Key Takeaways

Importance of Operations in different industries

Trade-offs in Operations (responsiveness, out-of-stock, costs, etc)

Key performance measures: *flow rate, cycle time, flow time and inventory.*

Process flow diagram as a tool to analyze processes

Methodology to calculate the process capacity and bottlenecks.

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